

Elvis Martis

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SUMMARY

Over 5 years of experience as a computational chemist specializing in molecular dynamics, free-energy calculations, structure-based drug design, and protein-protein interactions. I have rich experience with small molecule discovery focused on hit identification and lead optimization. Recently, I have started using state-of-the-art protein design algorithms like Rosetta and ProteinMPNN to find new peptides that disrupt protein-protein interactions. My background in pharmacy and computational chemistry gives me a unique edge in understanding the underlying biology of diseases and the ability to use this knowledge to design computational experiments that provide new insights or directions to aid drug design.

EXPERIENCE

US2B, Nantes Université | Postdoctoral Fellow **2023-Till date**

- *In-Silico* Screening of small molecule hits for protein-protein interfaces.
- Molecular Docking, Free Energy Calculations, structure-based drug design.
- Designing peptide binders that inhibit protein-protein interactions.

Bombay College of Pharmacy **Feb. 2020 – Sept. 2023**
Assistant Professor [ad-hoc] **Mumbai, India**

- Subjects Taught: Organic and Medicinal Chemistry
- Students Co-Supervised for Bachelor's and Master's projects

ChemBio Discovery Inc. (Boston, MA, USA.) **Remotely from India**
Computational Chemist **Feb. 2019 - Feb. 2020**

- Early-Phase Small molecule Discovery
- Lead optimisation (structure-based strategies)
- Molecular dynamics simulations and free energy calculations

EDUCATION

Bombay College of Pharmacy [Mumbai University] **Mumbai, India**
Ph.D. (Pharmaceutical Chemistry) **2015-2019**

- Studied the effect of mutations on proteins
- Developed a method to understand Drug-resistant mutations
- Molecular Docking, Free Energy Calculations, Essential dynamics of proteins.

Bombay College of Pharmacy [Mumbai University] **Mumbai, India**
M.Pharm (Pharmaceutical Chemistry) **2011-2013**

- Developed 3D-QSAR (CORIA) models for acetylcholinesterase.
- Molecular Docking, 3D-QSAR, Virtual Screening, *In-vitro* Assay.

VES College of Pharmacy [Mumbai University] **Mumbai, India**
Bachelor of Pharmacy **2007-2011**

- Foundation courses on therapeutic agents and their mechanisms.

SKILLS

Computational Chemistry: Molecular Dynamics Simulations, Structure-based Drug Design, Protein Structure Prediction, Free Energy Calculations, QSAR and 3D-QSAR.

Software suits: AMBER MD, Rosetta suite, Schrödinger (Small molecule suit), Osprey (Protein design), Haddock and Piper (Protein-protein docking), ORCA (QM calculations), RDKit (Chemoinformatics), ProteinMPNN (protein design).

Programming Languages: Python, BASH.

RECENT PUBLICATIONS

- Ten Quick Tips to Perform Meaningful and Reproducible Molecular Docking Calculations. Martis EAF, Téletchéa S. *Plos Computational Biology*, (**in production 2025**).
- Enhanced solubility and oral bioavailability of niclosamide-sulfobutylether- β -cyclodextrin sodium inclusion complexes: An approach to improve its repurposing potential. M Kulkarni, V Shirsat, E Martis, S Vibhute, T Patil, S Kulkarni, E Coutinho. *Materials Today Communications* 39, 109202 (**2024**).
- A-kinase anchoring proteins are enriched in the central pair microtubules of motile cilia in *Chlamydomonas reinhardtii*. VG Rao, AA Shendge, PP D’Gama, EAF Martis, S Mehta, EC Coutinho, *FEBS letters* 598 (4), 457-476 (**2024**).
- Identification of potential antileishmanial 1, 3-disubstituted-4-hydroxy-6-methylpyridin-2 (1H)-ones, in vitro metabolic stability, cytotoxicity and molecular modeling studies. MR Borkar, EAF Martis, S Nandan, RH Patil, A Shelar, KR Iyer, K Raikumar, et. al. *Chemico-Biological Interactions* 351, 109758 (**2022**).
- Design, synthesis and SAR of antitubercular benzylpiperazine ureas. S Satish, R Chitral, A Kori, B Sharma, J Puttur, AA Khan, D Desle, E Martis, et. al. *Molecular Diversity* 26, 73-96 (**2022**).

AWARDS

2020-21: Finalist for the BiosolveIT Scientific Challenge.

2012: Bagged the best research project at “Avishkar- Inter-University Research Convention 2012” at University Level (Medicine and Pharmacy, Post-Graduate Category).

2011: Best Outgoing Student of Vivekanand Education Society College of Pharmacy.

2010: Best Poster Award at 3rd Indian Pharmaceutical Association Student’s Congress.

FELLOWSHIPS & GRANTS

2015: Nvidia GPU seeding Grant awarded jointly with Prof Dr Evans Coutinho.

2015: Modelling Chemical and Biological (Re)activity 4 (MCBR4) (IndoGerman Consortium) visiting fellowship (Feb to April 2015) to HeinrichHeine-Universität Düsseldorf, Institut für Pharmazeutische und Medizinische Chemie, Germany under Prof Dr Holger Gohlke.

2014-16: BASF SE fellowship for doctoral studies.

2011-13: The government of India, Ministry of Human Resources’ scholarship for pursuing Master of Pharmacy.